CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY CHANGA

B.Tech. (CL/EC/ME) Sem - I Examination, 2009-10
EE - 101 Fundamentals of Electrical & Electronics Engineering

Date: 21.12.2009, Monday Time: 2:00p.m to 5:00p.m Maximum Marks: 70

Instructions:
1. Attempt all questions from both sections.
2. Use separate answer book for each sections.
3. Figures to the right indicate full marks.
4. Assume suitable additional data, if necessary.

SECTION I

Q-1 A. Derive expression of equivalent inductance when two magnetically coupled coils are connected in series in two different ways. **7**

B. A steel ring of 25 cm mean diameter and circular section of 3 cm diameter has a gap of 1.5mm length. It is wound uniformly with 700 turns of wire carrying current of 2 amp. Calculate: (1) M.M.F (2) Flux density (3) Magnetic flux (4) Reluctance. Take μ_r=200. **3**

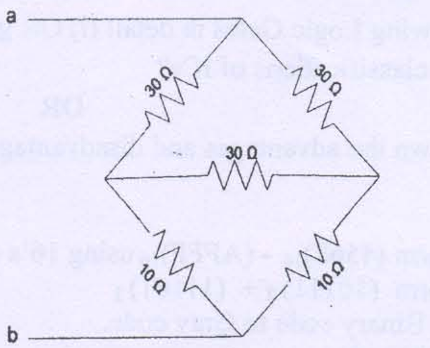
C. Define the following terms: **2**

1] Magnetic flux 3] Reluctance
2] M.M.F 4] Permeability

Q-2 A. Define temperature coefficient of resistance. Derive the following expression incase of good conducting material. Also state its significance. **7**

$$\alpha_{t2} = \frac{1}{\left(\frac{1}{\alpha_{t1}}\right) + (t_2 - t_1)}$$

B. Find out resistance between terminals a and b as shown in figure. **3**



C. State and explain Kirchoff's current law and kirchoff's voltage law. **2**

OR

Q-2 A. A capacitor and resistor are connected in series with dc source of 'V' volts. Derive an expression for the voltage across capacitor after 't' seconds during charging and discharging. **7**

B. A capacitor is charged with 5000 μC. If the energy stored is 1 joule. Find (a) voltage and (b) capacitance. **2**

C. Derive an expression of energy stored in capacitor of 'C' farad when voltage 'V' is applied across it. **3**

[P.T.O]

Q-3 A. Prove analytically that the current in a purely inductive circuit lags behind its voltage by 90°. And also prove that average power consumption in a pure inductor is zero when ac voltage is applied.

B. With graphical representation explain series R-L-C resonance. 2

C. Define following terms: (Any Four) 2

(a) Amplitude (c) Power Factor

(b) R.M.S value (d) Crest factor

OR

Q-3 A. Prove analytically that the current in a purely capacitive circuit leads its voltage by 90°. And also prove that average power consumption in a pure capacitor is zero when ac voltage is applied. 7

B. A capacitor of 10μF is connected across 230 V, 50 Hz supply. Find the capacitive reactance and the current drawn from the supply. What alternation in current drawn will occur if frequency is (1) doubled and (2) reduced to half? The supply voltage is kept constant in each case. 4

SECTION – II

Q-4 A. Draw and explain the *V-I* characteristics of PN junction. Also explain the operation of full-wave Bridge rectifier. 7

B. A crystal diode having internal resistance of r_f = 20Ω is used for half-wave rectification. If the applied voltage v = 50 sinwt and load resistance R_L= 800 Ω find :(i) I_m, I_{dc}, I_{rms} (ii) AC power input and DC power output (iii) DC output voltage (iv) efficiency of rectification 3

C. Explain the operation of Combination Clipper. 2

Q-5 A. Why voltage stabilizer circuit is required? How Zener diode is work as Voltage Regulator. 7

B. Explain following Logic Gates in detail (i) OR gate, (ii) EX-NOR gate. 2

C. Give various classifications of ICs? 2

OR

Q-5 A. (I) Write down the advantages and disadvantages of Digital system over the Analog system. 7

(II) (a) Perform $(456F)_{16}$ - $(AFFF)_{16}$ using 16's complement.

(b) Perform $(10111)_2 + (11101)_2$

B. Covert given Binary code to Gray code. 2

(i)1011, (ii)11111011

C. (i) Convert $(345)_{10}$ number into Hexa-decimal number. 2

(ii) Convert $(11010110111011)_2$ number into Octal number.

Q-6 A. Explain: how transistor uses as an amplifier? Derive the relationship between α and β. 7

B. Give definition of β, also derive the output current equation in Common Emitter configuration. 3

C. Only draw input and output characteristic curves of Common Emitter configuration 2

[P.T.O]

OR

Q-6 **A.** Explain in detail input and output characteristic curves of Common Base configuration. **6**

B. Give definition of α, also derive the output current equation in Common Base configuration. **3**

C. Find the α rating of the transistor shown in below figure. Hence determine the value of I_c using both α and β rating of the transistor. **3**